ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 4000 J	400 W x 10 s.
3	**(b)** algal bloom and fish kill	Eutrophication.
4	**(a)** the ground is heated by sunlight and warms the air	Surface heating.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** increase	More absorption.
7	**(b)** water filtration and flood control	Natural sponge.
8	**(b)** absorption of UV by ozone	Ozone heating.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Albedo is the fraction of sunlight reflected; snow reflects a large fraction, so its albedo is high.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but the composition of the atmosphere does not explain why CO_2 is a greenhouse gas.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorbed = 0.7 x 342 (1); = 239 W/m squared (1).	1+1
12	Long coastline and monsoon dependence (1); Large population in agriculture (1).	1+1
13	UV has the energy to break O2 and O3 bonds (1); Visible light has less energy (1).	1+1
14	Ice-albedo feedback (1); Release of methane from thawing permafrost (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1).	1+1+1
16	Dry air has little water vapour to trap heat (1); Quick cooling at night (1); Bare sand heats up fast (1).	1+1+1
17	30 units are reflected and 70 are absorbed (1); The Earth radiates back 70 units in equilibrium (1); Greenhouse gases delay the escape of heat (1).	1+1+1
	OR: Saving = 20 W x 100 = 2 kW (1); Energy = 2 x 8 x 250 = 4000 kWh (1); CO2 = 3200 kg (1).	1+1+1
18	Absorb CO2 and release O2 (1); Respiration uses much of the oxygen too (1); Better described as carbon stores and biodiversity hotspots (1).	1+1+1
	OR: Releases carbon stored for millions of years (1); Natural sinks cannot absorb all (1); CO2 accumulates (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hydrosphere (1). (ii) Atmosphere (1). (iii) Biosphere (1). (iv) Rain from the atmosphere (water vapour condenses) falls on the geosphere and flows into the hydrosphere (1); plants of the biosphere absorb water from the soil and return water vapour to the atmosphere (1).	1+1+1+2
20	Limestone heated: CaCO3 gives CaO + CO2 (1); Fuel burning adds CO2 (1); Cement 0.9 t CO2 per tonne (1); Recycling steel uses less energy (1); Renewable power and efficiency lower emissions (1).	1+1+1+1+1
	OR: Volume = 0.8 x 20 x 10^6 = 1.6 x 10^7 cubic m (1); Harvestable = 9.6 x 10^6 cubic m (1); More than the demand of 5 x 10^6 (1); Storage and recharge required (1); Reduces dependence on groundwater (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.

- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs $\mathrm{CO_2}$ and releases $\mathrm{O_2}$ (plants release $\mathrm{CO_2}$ only in respiration).
- Not converting units in rainwater problems: $\mathrm{area} \times \mathrm{depth}$ gives $\mathrm{m^3}$, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** its increase beyond natural levels enhances the greenhouse effect	Balance matters.
2	**(b)** weather and communication	Space technology.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 1.8 tonnes	12000 x 150 g.
5	**(c)** reducing the use of chemical fertiliser	Fertiliser production emits CO2.
6	**(b)** 750 s	d/c.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** 15 kWh	0.1 x 150.
9	**(c)** A is true but R is false.	A is true; the ozone layer is in the stratosphere, so R is false.
10	**(d)** A is false but R is true.	Plants absorb carbon dioxide in photosynthesis, so A is false; R is true: $6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The 100-year gas (1); Cuts in long-lived gases are urgent (1).	1+1
12	Water absorbs most radiation (1); Dark appearance (1).	1+1
13	Seasonal photosynthesis changes uptake (1); Rising trend is due to fossil fuels (1).	1+1
14	Net emission equals absorption (1); A community planting trees to offset emissions (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CO2 dissolves forming carbonic acid (1); pH of the sea falls (1); Harms corals and shellfish (1).	1+1+1
16	420 + 50 = 470 ppm (1); Increase 50/420 (1); = 11.9 per cent (1).	1+1+1
17	Reservoir that absorbs more carbon than it releases (1); Forests, oceans, soil (1); Carbon capture and storage (1).	1+1+1
	OR: Excess nitrate runs off (1); Eutrophication (1); Organic manure and crop rotation with legumes reduce use (1).	1+1+1
18	Land heats faster than the sea in summer (1); Moist sea air flows in and gives rain (1); Warmer oceans give more moisture and heavier but erratic rainfall (1).	1+1+1
	OR: Area needed to support a person's consumption (1); Compare by diet, travel, energy use (1); Reduction by LiFE habits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $100 \times 0.8 = 80$ m^3 (1). (ii) $80 \times 1000 = 80\,000$ L (1). (iii) Methane, nitrous oxide or water vapour (any one) (1). (iv) Greenhouse gases trap heat radiated from the Earth's surface (1), so less heat escapes to space and the average temperature of the atmosphere rises (1).	1+1+1+2
20	Two identical containers: black and white (1); Same light, same time (1); Measure temperature rise (1); Black rises more (1); Link: albedo of snow, forest, deserts (1).	1+1+1+1+1
	OR: Ice cores trap ancient air (1); Glacial: 180 ppm, cold (1); Pre-industrial: 280 ppm (1); Today: 420 ppm, faster change (1); Human activity as the cause (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.

- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs $\mathrm{CO_2}$ and releases $\mathrm{O_2}$ (plants release $\mathrm{CO_2}$ only in respiration).
- Not converting units in rainwater problems: $\mathrm{area} \times \mathrm{depth}$ gives $\mathrm{m^3}$, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 12 kg	15 kWh x 0.8.
3	**(a)** melting ice lowers albedo and increases warming	Amplifies the change.
4	**(b)** 4	A sphere has 4 times the area of its cross-section.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** 90 per cent	90/100.
7	**(b)** more CO2 helps plants grow and absorb more CO2	Dampens the change.
8	**(a)** 342 W/m squared	1370/4.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Clouds, rain and winds form where the water vapour is, in the troposphere.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but they are two different ways of nitrogen fixation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Advantages: absorbs CO2, restores habitat (1); Limitations: takes years, land needed (1).	1+1
12	44/12 x 1 (1); = 3.67 kg (1).	1+1
13	Warm water expands (1); Melting land ice adds water (1).	1+1
14	UV damages phytoplankton (1); Affects marine food chains and CO2 absorption (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	C + O2 gives CO2: 12 g gives 44 g (1); CO2 = 6 x 44/12 = 22 kg (1); Oxygen used = 16 kg (1).	1+1+1
16	CO2 dissolves in rainwater forming acid (1); Reacts with rocks (1); Carbonates carried to the sea (1).	1+1+1
17	Photosynthesis in oceans gives about half of the oxygen (1); They absorb CO2 (1); Base of the marine food chain (1).	1+1+1
	OR: Atmosphere is mostly transparent to visible light (1); Surface absorbs and warms (1); Surface heats the air by conduction and infrared (1).	1+1+1
18	CO2: fossil fuels, deforestation (1); CH4: paddy fields, cattle, landfills (1); Methane traps more heat per molecule but is short-lived (1).	1+1+1
	OR: Changes in weather patterns (1); Loss of polar bear habitat (1); New shipping routes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Carbon dioxide + water $\xrightarrow{\text{sunlight, chlorophyll}}$ glucose + oxygen, i.e. $6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$ (1). (ii) Respiration (1). (iii) Burning of fossil fuels, deforestation or burning of crop residue (any one) (1). (iv) Trees absorb carbon dioxide during photosynthesis and store the carbon in their bodies (1), so the carbon dioxide in the atmosphere reduces (1).	1+1+1+2
20	Ocean uptake = 3 Gt (1); Land uptake = 2.5 Gt (1); Atmosphere gains 4.5 Gt (1); Rise = 4.5/2.1 = 2.1 ppm (1); Shows the need to cut emissions (1).	1+1+1+1+1
	OR: Warm oceans expand and evaporate more (1); Atmosphere: stronger cyclones, changed rainfall (1); Geosphere: coastal erosion and flooding (1); Biosphere: coral bleaching, fish migration (1); Feedback to the atmosphere (1).	1+1+1+1+1

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